

Artificial Intelligence Fundamentals

CSC-113

Fall 2026 Section 2801 3 Credit(s) 08/17/2026 to 12/16/2026 Delivery Method: Traditional
Modified 08/18/2026

How & When our Course Meets

Course Format:

IN-PERSON

Meeting Days / Times / Location:

MONDAY-FRIDAY

11:15am - 12:40pm

Schedule Adjustment and Withdrawals

Each semester, you have a window of opportunity to make changes to your course schedule or withdraw. The timing of when you withdraw impacts how much, if any, of your tuition may be refunded and may also impact your financial aid.

To view the withdrawal deadlines, please visit [Wake Tech's Calendar](https://www.waketech.edu/student-services/registration-student-records) (<https://www.waketech.edu/student-services/registration-student-records>).

To view the comprehensive withdrawal policy, visit [Withdrawal Policy](https://www.waketech.edu/catalog/academic-information) (<https://www.waketech.edu/catalog/academic-information>).

To contact financial aid, visit [Financial Aid](https://www.waketech.edu/%EF%AC%81nancial-aid) (<https://www.waketech.edu/%EF%AC%81nancial-aid>).

Meet Your Instructor

Ethan T Gardner

Email: etgardner@waketech.edu

What This Course Is About

This course provides a survey of artificial intelligence and machine learning. Topics include the history, development, and current applications of artificial intelligence and machine learning. Upon completion, students should be able to demonstrate general artificial intelligence and machine learning concepts. Credit Hours: 3

What You'll Learn: Student Learning Outcomes

Upon completion of this course, students will be able to:

SLO 1

Understand the history and evolution of artificial intelligence.

SLO 2

Identify key concepts and terminology in AI and machine learning.

SLO 3

Recognize various applications of AI across different industries.

SLO 4

Explain the fundamental differences between AI approaches (symbolic AI, machine learning, deep learning).

SLO 5

Demonstrate awareness of ethical considerations in AI development and deployment.

SLO 6

Evaluate the potential impact of AI technologies on society and various professions.

What You'll Need: Textbook & Materials

Eagle Advantage is Wake Tech's textbook program for students in degree programs, excluding Career & College Promise (CCP) and Cooperative Innovative High School (Early College) students. The program ensures all required printed and digital textbooks are bundled and ready for pickup or delivery on or before the first day of class - saving you time and money and ensuring you're ready to learn on Day One.

If you are opted-in to Eagle Advantage, visit the 'Course Materials-BibliU' link in Blackboard for more information about the textbook details for your course. You can also find out more information by visiting [FAQs and Troubleshooting – BibliU \(https://support.bibliu.com/hc/en-us/sections/360004011038--FAQs-and-Troubleshootin\)](https://support.bibliu.com/hc/en-us/sections/360004011038--FAQs-and-Troubleshootin).

If you opted-out of Eagle Advantage, you are responsible for purchasing any of the required texts/materials for this course. For more information about Eagle Advantage, contact our [Bookstore](https://www.waketechcampusstore.com/SelectTermDept) (<https://www.waketechcampusstore.com/SelectTermDept>).

Artificial Intelligence Basics: A Non-Technical Introduction

Author: Tom Taulli

Publisher: Apress

Year: 2019

ISBN: 1484250273

Special Software or Hardware Requirements

No specific hardware requirements for this course. Students will primarily use standard web browsers and basic software applications to complete assignments. Any specialized tools or platforms will be accessible through college resources or free online services.

✓ How Your Work will be Evaluated: Grading Information

Criteria

Type	Weight	Topic	Notes
Discussion Board	20%		
Assignments	50%		
Final Project	30%		

Breakdown

This course will use a typical grading scale.

Grade	Range	Notes
A	90-100	
B	80-89	
C	70-79	
D	60-69	
F	0-59	

How You'll Proceed Through the Course: Course Calendar

Please follow the course schedule below.

When	Topic	Notes
Week 1-2	Introduction to AI	What is AI, Historical timeline of AI, Real-world AI applications, Impact of AI on society
Week 3-4	Basic Concepts of AI	Logic and problem-solving, Search algorithms (DFS, BFS), Simple puzzles to illustrate searching
Week 5-6	Knowledge Representation	Symbolic AI, semantic networks, Ontologies, frames, rules, Expert systems examples
Week 7-8	Introduction to Machine Learning	ML overview and importance, Supervised vs. unsupervised learning, Examples and applications
Week 9-10	Neural Networks Made Simple	Neural network fundamentals, Biological inspiration, Layers and data transformation, Visualization of simple networks
Week 11	Natural Language Processing	Basic NLP tasks, Chatbots and conversation systems, Real-world applications
Week 12	AI in Robotics and Computer Vision	Autonomous systems, Image classification, Object detection
Week 13	Ethics of AI	Ethical implications, Bias in data and models, Privacy concerns, Economic and social impacts
Week 14	Introduction to MCP topics; Final Project	Discuss how modern LLMs leverage external tools / data through an abstraction known as Model Context Protocol (MCP) servers Project kick-off, Group work on AI applications
Week 15	Project Presentations	Group presentations, Course wrap-up

* How I Will Help You Succeed: Instructor Policies & Procedures

Late Work Policy

Each project/assignment/exercise will come with specific instructions for completion. All studio/class work MUST be completed (if applicable)!

It is your responsibility throughout the semester to keep up with your course work. It is not the responsibility of the instructor to advise you of missed work during the term or on the last week of the term.

Exams must be taken on the assigned date; otherwise a grade of zero will be assigned. Any exception to this policy must be the result of a documented emergency or prior arrangement with the instructor. There will be no exceptions to this rule. ANY MAKEUP EXAMS must be taken prior to the scheduled exam date. This must be pre-arranged with the instructor. Last minute arrangements are not acceptable.

Statement on AI Usage

Students are encouraged to utilize AI tools and resources as part of their coursework, unless expressly prohibited for specific assignments or activities. When using AI tools, students must properly cite and attribute AI-generated content in their work. For this course, students should document:

1. The AI tool used (e.g., ChatGPT, Claude, Bard)
2. The prompt provided to the AI
3. How the AI-generated content was incorporated or modified

This documentation should be included as an appendix or footnote in assignments where AI tools were utilized. The goal is to promote responsible and transparent use of AI technologies while maintaining academic integrity.

How the Department Will Help You Succeed: Department Policies & Procedures

Tutoring and Learning Center

The Tutoring and Learning Center (TLC) offers Wake Tech students free tutoring and learning support for their classes. The mission of the TLC is to help students understand what they are learning in class, develop the skills they need to be successful in college and the workplace, and empower them to be independent learners.

Visit the TLC at [Tutoring and Learning Center Site](https://www.waketech.edu/student-services/tutoring-learning-center) (<https://www.waketech.edu/student-services/tutoring-learning-center>)

Information & Digital Technology Care Team

The IT Care Team provides specialized advising and coaching support for University Transfer students interested in IT and Associate in Applied Science students enrolled in Information Technology programs.

Visit the IT Care Team website at [IT Care Team Site](https://www.waketech.edu/student-services/advising/information-technology) (<https://www.waketech.edu/student-services/advising/information-technology>)

How the Division Will Help You Succeed: Division Policies & Procedures

Process for Expressing Concerns about the Course

1. Please discuss concerns with your instructor first.
2. If your concerns are not resolved, you may contact the Department Head or Program Director of the department.

General Division Questions

For general questions regarding the Information Technology Division or assistance with program resources, students may contact Associate Dean Emmie Mercer at ermerc1@waketech.edu for guidance or referral to the appropriate department.

Information Technology Technical Standards

This page outlines the technical standards and essential skills expected of students in Information Technology programs. Review this page for descriptions of the critical-thinking, communication, interpersonal, behavioral, and technical abilities needed to complete coursework and perform effectively in professional IT environments: [IT Technical Standards](https://www.waketech.edu/about-wake-tech/divisions/information-technology/technical-standards) (<https://www.waketech.edu/about-wake-tech/divisions/information-technology/technical-standards>).

How the College Will Help You Succeed: Wake Tech Services, Policies and Procedures

Wake Tech is committed to supporting you by providing the resources needed to succeed. This is a brief overview of support available at the College. You can always contact me to help you make the right connection with a college resource. We are here to help!

Student Support

As a Wake Tech student, you have free access to non-academic and academic resources to help you with a range of challenges. These resources include academic services, such as advising, tutoring and academic coaching to help you understand your career and academic pathway, Disability Support Services, Tutoring and Learning Centers, Career Services, and Library Services. The college also offers assistance with basic life needs through the Care Center. These services include mental health resources, emergency financial assistance, safe housing, transportation, and technology. If you find yourself in need, you can reach out to any faculty member or visit this website [Student Support](https://www.waketech.edu/studentsupport) (<https://www.waketech.edu/studentsupport>) to learn more about the help that we can provide.

Student Activities

Wake Tech is continually developing new ways for you to get involved with the College! For information about the Student Centers, Student Government Association, Clubs and Organizations, Athletics, and more, please visit the [College Catalog \(https://www.waketech.edu/catalog/student-services-and-activities\)](https://www.waketech.edu/catalog/student-services-and-activities).

Technical Support

Wake Tech has several technology platforms to help you succeed. You may find that technical questions come up regarding your use of technology and Wake Tech is here to help. For technical support, including Blackboard, WTCC email, MS Teams, and Self Service visit the [IT Helpdesk page \(https://www.waketech.edu/help-center/its/topics\)](https://www.waketech.edu/help-center/its/topics).

Student Code of Conduct, Rights, and Responsibilities

You may have questions about your rights and responsibilities as a student. Wake Tech has a range of policies to address such concerns. For information about the Academic Integrity Policy, Complaints, Title IX Policy, Student Appeals, and more, please visit the [College Catalog \(https://www.waketech.edu/catalog/student-code-conduct-rights-and-responsibilities\)](https://www.waketech.edu/catalog/student-code-conduct-rights-and-responsibilities).

Academic Policies

There are several Wake Tech policies related to academics that you might need to reference in order to succeed in your classes. If you have questions about the Attendance Policy, Student Email Policy, Absences for Religious Observances, Withdrawal Policies, Grades, and more, please visit the [College Catalog \(https://www.waketech.edu/catalog/academic-information\)](https://www.waketech.edu/catalog/academic-information).

Student Information Privacy

Wake Tech is committed to protecting your privacy. The Family Educational Rights and Privacy Act (FERPA) is a federal law enacted in 1974 that protects the privacy of student education records. To find out more about College policies and procedures that support your privacy, please visit the [College Catalog \(https://www.waketech.edu/catalog/registration-and-student-records\)](https://www.waketech.edu/catalog/registration-and-student-records).

Campus Policies

There are several Wake Tech policies related to campus procedures that you might need to reference to feel comfortable on campus. For information about College Police and Safety, the Acceptable Use of College Resources Policy, (Student Dress, Pets, Visitors and Children on Campus, and more, please visit the [College Catalog \(https://www.waketech.edu/catalog/campus-policies-and-procedures\)](https://www.waketech.edu/catalog/campus-policies-and-procedures).

Registration & Student Records

For information about the Class Schedules, Registration Dates, Transcript Requests, Transfer Credits, Program of Study Changes, and more, please visit the [College Catalog \(https://www.waketech.edu/catalog/registration-and-student-records\)](https://www.waketech.edu/catalog/registration-and-student-records).

Financial Aid

Wake Tech is here to answer your financial aid questions. For information about Enrollment Requirements, Grants, Loans, Scholarships, and more, please visit the [College Catalog](https://www.waketech.edu/catalog/financial-aid) (<https://www.waketech.edu/catalog/financial-aid>).

Inclement Weather/Wake Tech WARN

North Carolina can have some unpredictable weather. For up-to-date information about Emergency Closings, please sign up for the College alert system, [Wake Tech WARN](https://www.waketech.edu/about-wake-tech/administrative-offices/campus-police/waketech-warn) (<https://www.waketech.edu/about-wake-tech/administrative-offices/campus-police/waketech-warn>). For more information about the inclement weather policy, please visit the [College Catalog](https://www.waketech.edu/catalog/campus-policies-and-procedures) (<https://www.waketech.edu/catalog/campus-policies-and-procedures>).

Additional Items

Course Requirements and/or Expectations

1. Students are expected to be active in course discussions.
2. Reading lecture material is paramount to success in the course.
3. Communication with the professor is vital if you are experiencing issues or need help/clarification.
4. Students will have to demonstrate understanding of AI concepts through assignments, discussions, and assessments.

Wake Tech Core Values

Wake Technical Community College will structure its operations, training and educational programs around the Core Values of accountability, respect, responsibility, critical thinking, communication, and collaboration.

- **Accountability**—Accountability is essential for an environment of learning. Those who are accountable stand by their words and actions, taking full responsibility for what they create and for what they contribute to the community.
- **Respect**—Respect is a prerequisite for enhancing learning. Community members who respect themselves and others help create a safe, yet open, climate of learning.
- **Responsibility**—Responsibility is the root of success. Students who assume personal responsibility for their education will reach their goals. Responsible students also make contributions to their communities.
- **Critical Thinking**—Critical thinking is the fundamental purpose of higher education. The ability to solve problems through the application of the appropriate skills is critical to all disciplines.
- **Communication**—Communication is increasingly the key competency for living and working in the information age. Communicating effectively in oral and written forms through traditional and new media is a powerful tool for personal and career success.
- **Collaboration**—Collaboration, by bringing together individual knowledge and talents, creates teams that are greater than the sum of their parts. Such teamwork maximizes benefits to individuals and

the community.